

BONAIRE INTL, Bonaire, Sint Eustatius and SabaWMO: **789900**Lat: **12.131N**Lon: **68.269W**Elev: **7**StdP: **101.24**Time Zone: **-4.00 (W04)**Period: **09-19**WBAN: **00413****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.2	23.9	19.6	14.4	26.6	20.1	14.8	26.5	12.0	28.7	11.4	28.5	3.3	80	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.9	32.6	26.7	32.2	26.5	31.9	26.4	27.7	30.8	27.2	30.7	26.9	30.6	6.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	29.9	26.2	21.6	29.6	26.0	21.4	29.6	88.1	31.3	86.3	31.1	85.0	30.6	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.7	10.2	21.7	34.0	0.7	0.8	21.2	34.5	20.7	35.0	20.3	35.4	19.8	36.0		
(5)			20.4	29.1	0.7	2.0	19.9	30.6	19.4	31.8	19.0	32.9	18.5	34.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	27.1	27.1	27.6	28.2	28.8	29.1	29.1	29.6	29.7	29.4	28.6	27.7
	DBStd	1.19	0.77	0.74	0.75	0.69	0.62	0.57	0.62	0.67	0.89	0.95	1.11	0.90
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6757	532	479	546	547	583	573	591	607	591	601	557	550
	CDD18.3	3715	273	246	287	297	325	323	333	349	341	342	307	292
	CDH23.3	42029	2575	2261	2842	3169	3734	3851	3982	4366	4374	4244	3561	3071
	CDH26.7	14689	535	473	685	909	1283	1461	1514	1895	2005	1804	1297	827
Wind	WSAvg	6.6	6.5	6.9	7.0	7.1	7.6	7.8	7.3	6.7	6.1	5.6	4.9	6.3
Precipitation	PrecAvg	527	47	20	14	18	15	19	31	22	40	74	141	84
	PrecMax	1006	147	90	76	95	82	88	113	73	203	240	416	311
	PrecMin	268	2	0	0	1	0	1	7	1	3	2	36	5
	PrecStd	198	33	20	19	22	22	20	26	19	42	61	88	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	30.8	31.0	31.2	32.0	32.2	32.2	33.0	33.2	32.9	32.2	31.2
		MCWB	25.0	25.5	25.0	25.9	26.6	26.8	26.5	26.7	27.0	26.5	26.7	25.8
	2%	DB	29.9	29.9	30.2	30.9	31.2	31.5	31.5	32.2	32.7	32.2	31.8	30.6
		MCWB	24.7	24.2	24.7	25.6	26.1	26.2	26.2	26.4	26.7	26.4	26.4	25.6
	5%	DB	29.3	29.2	30.0	30.4	31.0	31.1	31.1	32.0	32.2	32.0	31.2	30.1
		MCWB	24.4	24.0	24.6	25.2	26.0	26.1	26.1	26.3	26.5	26.4	26.2	25.4
	10%	DB	29.0	29.0	29.4	30.0	30.5	30.9	30.9	31.5	31.9	31.3	30.9	29.9
		MCWB	24.3	24.0	24.3	25.0	25.7	26.0	26.0	26.3	26.5	26.4	26.1	25.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.1	26.1	27.1	27.2	27.9	28.3	28.7	28.2	27.9	27.4	26.7
		MCDB	29.0	29.7	29.5	30.2	30.8	30.5	31.0	31.9	31.0	30.9	30.8	30.2
	2%	WB	25.6	25.3	25.5	26.5	26.7	26.7	26.7	27.2	27.6	27.3	26.8	26.3
		MCDB	28.7	28.7	29.0	29.6	30.2	30.4	30.3	30.8	30.8	30.6	30.5	29.7
	5%	WB	25.2	24.7	25.2	26.1	26.4	26.6	26.5	26.7	27.2	27.0	26.6	25.9
		MCDB	28.5	28.2	28.7	29.2	29.8	30.2	30.1	30.5	30.7	30.4	30.1	29.1
	10%	WB	24.7	24.4	24.8	25.6	26.1	26.2	26.2	26.6	26.9	26.7	26.3	25.6
		MCDB	28.1	27.9	28.4	28.9	29.4	29.8	29.7	30.2	30.5	30.1	29.7	28.8
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	4.6	4.3	4.1	4.1	4.3	4.6	4.9	4.6	4.8	4.6
		MCDBR	4.6	4.8	4.7	4.5	4.3	4.3	4.5	4.8	4.9	4.8	4.9	4.7
		MCWBR	2.3	2.0	1.9	1.8	1.9	2.0	2.2	2.2	2.2	2.0	2.2	2.4
	5% WB	MCDBR	4.5	4.3	4.5	4.2	4.3	4.2	4.4	4.7	4.9	4.6	4.8	4.5
		MCWBR	2.6	2.2	2.3	2.1	2.2	2.4	2.5	2.4	2.5	2.2	2.3	2.5
Clear-Sky Solar Irradiance	taub	0.387	0.409	0.420	0.486	0.476	0.505	0.484	0.441	0.426	0.402	0.392	0.395	
	taud	2.554	2.468	2.446	2.238	2.301	2.199	2.279	2.440	2.474	2.574	2.589	2.559	
	Ebn at Noon	917	907	900	833	827	793	813	859	881	903	906	900	
	Edh at Noon	101	113	118	145	133	145	135	116	113	101	97	99	
All-Sky Solar Radiation	RadAvg	5.30	5.88	6.22	6.24	6.18	6.29	6.39	6.50	6.26	5.58	4.98	4.95	
	RadStd	0.21	0.18	0.31	0.24	0.27	0.24	0.18	0.26	0.24	0.30	0.50	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	-0.33	-0.24	-0.48	-0.48	N/A	N/A	N/A	N/A

Nomenclature: See separate page